PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

To:
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PCT

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY

(PCT Rule 43*bis*.1)

Date of mailing *(day/month/year)*	19-05-2025
Applicant's or agent's file reference 1403092801	**FOR FURTHER ACTION** See paragraph 2 below

International application No.	International filing date *(day/month/year)*	Priority date *(day/month/year)*
PCT/IB2024/062818	18-12-2024	18-12-2024

International Patent Classification (IPC) or both national classification and IPC
A61K47/36,A61K31/02,C08L5/04,C08L5/08,B82Y5/00,A61P35/00 Version=2025.01

Applicant
AGHAZADEH, HAMED,KAVA, ALI, and others.

1. This opinion contains indications relating to the following items:

- [x] Box No. I Basis of the opinion
- [ ] Box No. II Priority
- [x] Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- [ ] Box No. IV Lack of unity of invention
- [x] Box No. V Reasoned statement under Rule 43*bis*.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- [ ] Box No. VI Certain documents cited
- [ ] Box No. VII Certain defects in the international application
- [x] Box No. VIII Certain observations on the international application

2. **FURTHER ACTION**

If a demand for international preliminary examination is made, this opinion will be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1*bis*(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA/	Date of completion of this opinion	Authorized officer
Indian Patent Office Plot No. 32, Sector 14, Dwarka,New Delhi-110075 Facsimile No.	19-05-2025	Dr. Sateesh Kumar Meena Telephone No. +91-1125300200

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Box No. I Basis of this opinion

1. With regard to the **language**, this opinion has been established on the basis of:

   ☒ the international application in the language in which it was filed.

   ☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).

2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43*bis*.1(b)).

3. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:

   a. ☐ forming part of the international application as filed.

   b. ☐ furnished subsequent to the international filing date for the purposes of international search (Rule 13*ter*.1(a)),

      ☐ accompanied by a statement to the effect that the sequence listing does not go beyond the disclosure in the international application as filed.

4. ☐ With regard to any nucleotide and/or amino acid sequence disclosed in the international application, this opinion has been established to the extent that a meaningful opinion could be formed without a WIPO Standard ST.26 compliant sequence listing.

5. Additional comments:

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Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability

The questions whether the claimed invention appears to be novel, to involve an inventive step (to be non obvious), or to be industrially applicable have not been examined in respect of:

☐ the entire international application.

☒ claims Nos. 29-44

because:

☒ the said international application, or the said claims Nos. 29-44 relate to the following subject matter which does not require an international search *(specify)*:

The subject matter of claims 29-44 relates to a method for the treatment of the human or animal body, which does not require an international search by the International Searching Authority in accordance with PCT Article 17(2)(a)(i) and [Rule 39.1(iv)].

☐ the description, claims or drawings *(indicate particular elements below)* or said claims Nos. _____ are so unclear that no meaningful opinion could be formed *(specify)*:

☐ the claims, or said claims Nos. _____ are so inadequately supported by the description that no meaningful opinion could be formed *(specify)*:

☒ no international search report has been established for said claims Nos. 29-44

☐ a meaningful opinion could not be formed without the sequence listing; the applicant did not, within the prescribed time limit:

☐ furnish a sequence listing complying with WIPO Standard ST.26, and such listing was not available to the International Searching Authority in the form, language and manner acceptable to it.

☐ pay the required late furnishing fee for the furnishing of a sequence listing in response to an invitation under Rule 13*ter*.1(a).

☐ See Supplemental Box for further details.

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Box No. V **Reasoned statement under Rule 43*bis*.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement**

1. Statement

Novelty (N)	Claims	1-28	YES
	Claims	None	NO
Inventive step (IS)	Claims	None	YES
	Claims	1-28	NO
Industrial applicability (IA)	Claims	1-28	YES
	Claims		NO

2. Citations and explanations:

Reference is being made to the following documents:

D1:IN202311081092 A; (INDIAN INSTITUTE OF TECHNOLOGY BHU VARANASI [IN]); 13 September 2024

D2:WO2013075186 A1; (UNICAMP [BR]); 30 May 2013

D3:CN105963702 A; (UNIV HUBEI); 28 September 2016

D4:S Bhunchu et al.; Biopolymeric alginate-chitosan nanoparticles as drug delivery carriers for cancer therapy; Pharmazie 2014 Aug;69(8); 563-70.

D5:Yuhan Zhou et al.; Secretions from hypochlorous acid-treated tumor cells delivered in a melittin hydrogel potentiate cancer immunotherapy; Bioact Mater. 2021 Jul 24(9); 541-553.

The present invention relates to a nanomedicine product that can release HOCI (hypochlorous acid)at the tumor site in a controlled manner to treat lung cancer. HOCI is a potent antimicrobialand antitumor agent that can target tumor tissues while minimizing damage to healthy tissues.The produced nanoparticles have a two-layer structure: the inner layer contains alginate, PFP(perfluoropentane), and Ca(0C1)2 to release HOC1, and the outer layer has a chitosan coatingwith folate and salicylic acid for targeting and proton release.

Novelty under Article 33(2) of PCT:

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Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

The application does not meet the requirements of Article 6 PCT, because the claim 1 are not clear for the following reasons: Drafting of claims is not correct, because of the nature & the scope of the alleged invention can not be clearly identified. Claim 1 is claiming a nanomedicine product that can release HOCI (hypochlorous acid)at the tumor site in a controlled manner to treat lung cancer which is a product but subsequently claims are drafted in terms of functional steps rather essential product features. Product claims must have essential product features instead functional steps of making that product.The statement of claims should, therefore, be revised and all essential features of the invention should bring in the principal claim while the invention of subsidiary features dependent claims in the claim. Subsequent claims should elaborate on the features of the principal claim after the phrase "as claimed in claim ... wherein"

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.
Continuation of:

Continuation of Citation and Explanation(Box5)

None of the cited documents specifically disclose a A nonomedicine is developed using HOCl-PFp Alginate-chitosan nanoparticles to deliver HoCl to the tumor site, causing apoptosis in A549 cells and reducing lung cancer tumor growth with ultrasound irradiation. Therefore, the subject matter of claims 1-28 is novel.

Inventive step under Article 33(3) :-

The subject matter of claims 1-28 fails to meet the requirements of PCT Article 33(3) because it does not involve an inventive step in view of the disclosure of documents D1-D5.

D1 relates to a nano-formulation for management of lung cancer and a method of preparing the same, having the chitosan nanoparticles functionalized with an EGFR antibody and sialic acid for the delivering the potent chemotherapeutic drug, that is gemcitabine over-expressing non-small-cell lung- cancer (NSCLC) (A-549) cells. Further, the nano-formulation provides a pH-dependent controlled release of the chemotherapeutic drug in a targeted fashion with improved cellular internalization, physiochemical characterization and induces cytotoxicity, apoptosis and necrosis in A-549 cells through receptor-mediated approach. The chemotherapeutic agent is loaded on the nano-carrier for sustained release of the chemotherapeutic agent (see abstract; claims).
The difference between D1 and the current invention is use of core chemotherapeutic agent. While D1 is using gemcitabine, the claimed invention is using HOCI (hypochlorous acid). It is evident that use of HOCI to treat infection is not new and HOCl is morphing from a well-accepted disinfection chemical component to an increasingly used in vivo agent for applications in wound care and the naso-pharynx.

D2 describes nanoparticles with polymer walls as a carrier system coated with chitosan folate and containing a chemotherapeutic drug, with improved targeting of action sites, bioadhesion and biocompatibility, and less toxicity. The subject matter of the present invention can therefore be of great interest in sustained drug release systems used in the treatment of solid tumours orally, intravenously or topically (see abstract; claims).

D3 relates to the technical field of medicines, and in particular relates to a drug delivery controlled-release system having a tumor induced targeting capacity and a preparation method of the drug delivery controlled-release system.The drug delivery controlled-release system has three layers of core-shell structures, wherein the inner layer is drug-loaded meso-porous

silicon nanoparticles, the middle layer is folic acid modified azidation chitosan and the outer layer is polyethylene glycol. The preparation method of the drug delivery controlled-release system comprises the following steps: preparing the thiol functional meso-porous silicon nanoparticles; conducting double-sulfur functionalization; conducting alkynyl modification; removing a template; preparing the folic acid modified azidation chitosan; covering the chitosan; conducting PEG (polyethylene glycol) modification, and the like (see abstract; claims).

D4 is all about an overview of drug delivery systems for cancer treatment, describe the use of biopolymeric alginate–chitosan nanoparticles for anticancer drug delivery, and discuss the important characteristics of these nanoparticles for use in drug delivery.. Nanoparticulate drug delivery systems enhance cancer treatment by direct entry of nanometer particles into the fenestration in the vasculature of cancer cells. Nanoparticles for encapsulation of anticancer drugs are preferably prepared using natural polymers as carriers, with polysaccharides being particularly favorable. Alginate and chitosan polysaccharides have been widely used in nanoparticulate drug delivery systems because of their biodegradable, biocompatible, non-toxic and bioadhesive properties (see abstract).

D5 is all about a multifunctional HOCl-CDS hydrogel by loading the strongly immunogenic HOCl-CDS into a MELR hydrogel. HOCl-CDS hydrogel showed potent antitumor efficacy in melanoma by inducing direct tumor cell death and remodeling the TME. Furthermore, the HOCl-CDS hydrogel augmented the antitumor effects of anti-PD-1 treatment in the syngeneic melanoma mouse model, suggesting that HOCl-CDS hydrogel is a promising immunotherapeutic approach for local tumor treatments (see abstract).

Though D2, D3 & D4 are not focusing on HOCI based tumor treatment but concept of using any chemotherapeutic agent, sustained release or delayed release of any agent via desined drug delivery system remains the same. As far as the HOCI is concerned, in biological systems, HOCl can be regarded as the most effective and most common oxidizing agent. It is one of the foremost endogenous molecules for effective phagocytosis of invading bacteria or viruses. Naturally produced by the innate immune system, HOCl has known antimicrobial properties that also break down biofilm. It has also been shown to be anti-pruritic and anti-inflammatory, as well as to increase oxygenation to wound sites. It is recognized that HOCl can kill pathogens through its strong oxidizing potential. This cytotoxic oxidant also induces significant increases in cellular damage, as reflected in lipid peroxidation, protein oxidation, and DNA damage. Many chronic inflammatory conditions are associated with an increased risk of cancer development.

Hence, the combined disclosure of D1-D5 makes the claims 1-28 of present application obvious for a person skilled in the art and hence does not involve any inventive step

Industrial Applicability under PCT Article 33(4):

The subject matter of claims 1-28 is considered industrially applicable and meets the requirements of Article 33(4) PCT.